AI-Based Ballpark Quotation Tool

Automating R&D Cost Estimation for IVECO/FPT Industrial

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CHALLANGE & PROJECT OBJECTIVE

IVECO Group

IVECO Group is a global leader in commercial vehicles and industrial equipment.

Among its business units is **FPT Industrial** – Powertrain Technologies, responsible for developing engines and powertrain systems for on-road and off-road applications.

During the product development cycle, customers submit **Product Requests (PRs)** to propose new features or modifications. Each PR requires a quick, preliminary R&D effort estimation to assess feasibility and business value.

Problem Statement

Manual Intensity: Analyzing PR Excel files is slow and requires deep expert knowledge.

Fragmented Data: There is no centralized database linking historical PRs with their final R&D Offers.

HIGH DEPENDENCY: The process relies heavily on individual expert knowledge, creating a risk of knowledge loss.

Inconsistency: Different Customer Managers provide different quotes for similar requests.

Product Request

Input: Excel File

PRODUCT REQUEST		PR 21031 Rev G
Title: CWL New Model 100 hp		
Platform: CWL	**Plant: PLANT-LE**	
Engine: E3F6	**Tier: STAGE V**	

Vehicle Models:
Mac CE C.WhL

Description:
Mac CE C.WhL: E3F6, STAGE V, Boosted Curve : Rated Power : 71.9 kw @ 2200 RPM, Peak Power : 83.8 kw @ 2000 RPM, Max Torque : 453 Nm @ 1400 RPM
Mac CE C.WhL with E3F6 3.6L SCR-T 72kW Stage V engine with related compliant ATS (same concept of Mac CE T.LB in term of rating, ATS as Mac AG SPE vehicle) for Europe.

Mac CE C.WhL E3F6 3.6L SCR-T 72kW Tier4b engine with related compliant ATS (same concept of Mac CE T.LB in term of rating, ATS as Mac AG SPE vehicle) for NAFTA.

Engine controller will be ECU1. Dataset configured at 500kbps (no auto-baud rate) both for Stage V and T4B.

Replace oil Sump from Mac CE T.LB version to current CWL

Because of the necessities to have a dedicated PTO for emergency steering we need to adopt a

solution similar to SPE1 Tractors trought engine gearbox.

The change require to remove current oil fill tube and relocate it as per SSL solution with:
tube; Plug

The current oil fill tube will be plugged by COMPANY . CUSTOMER will fitup on EU units the Adaptor and pump for emergency steering

The change is required for stage V engine but the same modification can be extended to tier4B in order to manage one engine hardware.

Other change required is to relocate the relief valve currently installed on head cover where will be placed oil fill tube.

CUSTOMER officiallized investments for metal RACKs to support engines logistic transportation. Engine price should be revise consequently (no wooden pallet)

Updated Volumes + Added 1 DU

Evaluate the possibility to have one engine + ATS only, Stage V version, for EPA and ECE homologation on both EU and NA market. The volumes will remain the same

COUNTRIES SOLD TO: Europe, North America, ANZ (Australia, New Zeland)

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Project Objective

- Develop an AI-based tool for fast ballpark R&D estimations
- Automatically extract key information from PR Excel files
- Use historical PR–Offer pairs as training data
- Predict **function-level** R&D effort breakdown
- Allow Customer Managers and Function Owners to refine results

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Proposed Quote

Function Level R&D

Engineering Activity Summary and R&D expenses forecast (PE.02)

PE Function		Program Main Activities Description	Effort [hrs] Manpower	Effort [hrs] Bench	Effort [hrs] Vehicle	K€
Project Management		• Activity tracking and deliverable readiness	1170			100
Design	Base Design	• Release 4 new p/n of engine in PRP new drawings of engine (eVGT and WG), new flywheel, new exhaust flap orientation, new wiring harness (2), new fan pulley (1,4) ratio • Basic tech for ATS CFD, vibration and torsional analysis, verification for front PTO and calculation for 2 front end	2500 950			83
	ATS	• Installation checks of ATS and sensors ; assessment of full exhaust line, Kit ATS release, fluids analysis	840			
	EMS OBD	• Analysis of E/E system architecture and customer requirements according to Functional Safety approach • Verification of vehicle Interface functions • OBD verification according to Stage V / Tier4B, support for OBD verification on bench, support for field test	1600 960			20 48
Bench	Dev & Rel	• Calibration development for top power specific rating with specific base combustion with 1 HW of engine and 1 Kit ATS OBD verification • E15x1 overload (gamma) • E2 (thermal shock) • E39 test (gamma) • E46 test (gamma) • E75 test (gamma) • DF test • Homologation tests for USA • Homologation tests for EU	1080 160 600 1100 250 500 2300 160 240	1080 500 1800 3300 750 1500 4000 160 240		460 66 239 438 99 199 812 50 80
Application		• QG readiness and dataset release, (4 rating) calibration optimization on 2 vehicles, installation and functional checks (application sign off) / field test support →mid light classification top rating, 1 light and 2 super light • Dataset release	8800			194 18
Supplier R&D		• SupplierB related functions calibration, installation and functional verification on machine for DeNox and FIE • ATS Canning skin temperature for different layout, shaker test for new top rating, release drawing and validation for new DOC and SCROF				150 160
Technical Certification		• Managing official test for Homologation activities for EU/USA (1 parent)	400			30
Materials & Travels		• Materials • travels				40 10
					TOTAL	3.296

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Value Proposition

- **For the Customer Manager:**
  - **Speed:** Instant preliminary estimations replace days of manual analysis.
  - **Accuracy:** Data-driven predictions improve consistency across all quotations.
  - **Efficiency:** Drastic reduction in manual workload, allowing focus on high-value tasks

- **For the Business:**
  - **Agility:** Faster "Go/No-Go" decisions accelerate the sales cycle.
  - **Standardization:** A unified process across all functions and departments

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Sustainable Development Goal

- **SDG 9: Industry, Innovation and Infrastructure**
  - *Enhancing industrial capability through digitalization and AI-driven automation.*

- **SDG 12: Responsible Consumption and Production**
  - *Reducing "Engineering Waste" by minimizing duplicated efforts and optimizing resource planning*

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LLM Based Pipeline and Interface for Tool

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RESEARCH QUESTIONS

Research Questions

- **RQ1 -** How the tool transform raw text into activity type and activity effort?
- **RQ2 -** The tool find PR similarities: can we measure the accuracy of the search engine?
- **RQ3 -** Is a standalone regression model sufficient for business decision-making?
- **RQ4 -** How much tool is good in predicting activity type?
- **RQ5 -** How much tool is good in estimating effort intensity (hrs)?
- **RQ6 -** How much tool is good in estimating cost?

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Methodology

RQ: 1

Methodology

From Text to Estimation Content

- **Data Ingestion (Text to Features):**
  - Input: Raw Product Request (Excel/pptx).
  - Parsing: LLM extracts 27 structured features.
  - Transformation: Categorical data is encoded; numerical data is normalized.

- **Hybrid Processing Engine:**
  - Path A (Prediction): HCQE Model predicts Effort Intensity & Cost based on feature complexity.
  - Path B (Context): RAG Engine retrieves historical analogies.

- **Output:** DeepSeek Agent synthesizes the prediction with historical context into a business report

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RQ: 2

RAG System

Contextualized Cost Justification

Beyond Numerical Guesses

Our AI tool doesn't just provide cost figures; it offers depth and justification, moving beyond mere numerical predictions.

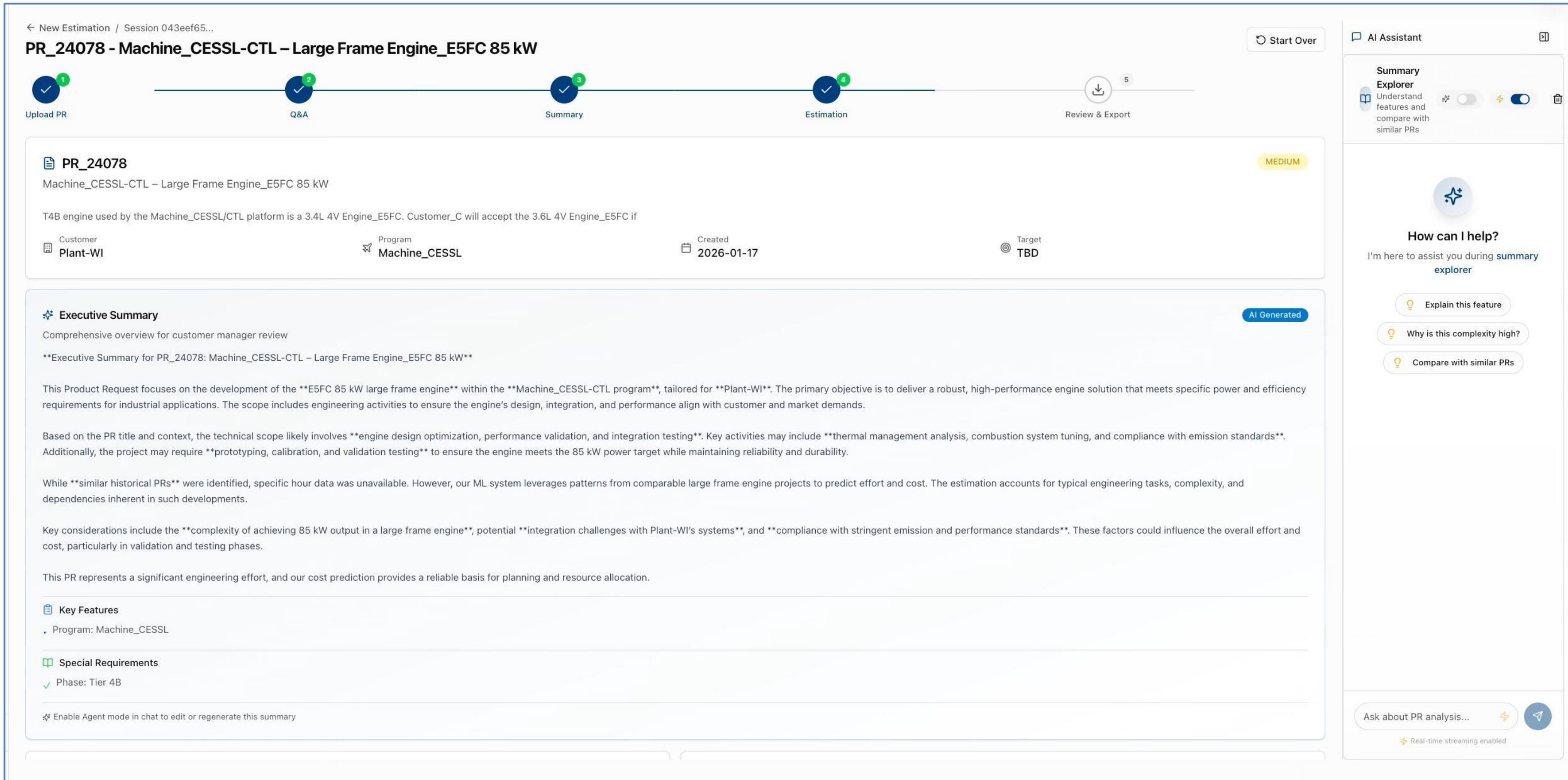
Retrieval from Qdrant

The Retrieval Augmented Generation (RAG) system actively searches and retrieves the top-3 most similar 'Reference Projects' from our Qdrant vector store.

Providing Context & Confidence

These retrieved projects provide invaluable historical context, enabling the AI to justify the estimated cost and enhance stakeholder confidence.

Summarized Input with LLM



The Analyst Agent

DeepSeek V3 Reasoning Layer

- **Role:** Synthesis & Reasoning (not just numerical output)
- **Input:** ML predictions + RAG context + historical patterns
- **Processing:** DeepSeek V3 analyzes and synthesizes all signals
- **Output:** Business-ready report explaining "Why" the cost is high
- **Intelligence Features:**
  - Flags risks and anomalies (e.g., "Calibration hours 2x historical average")
  - Provides confidence intervals and uncertainty bounds
  - Generates executive summary with key drivers
  - Recommends mitigation strategies for high-risk areas
- **Value:** Transforms raw predictions into actionable business intelligence

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RQ: 3-6

Methodology

The Mathematical Engine: HCQE & Multi-Tasking

- **Effort Intensity Prediction (Hours)**

  Since Cost ≈ Hours × Rate, our Cost accuracy reflects Effort accuracy.

  Key Driver: Manpower Hours has a 0.918 correlation with the target, proving the tool effectively captures effort intensity.

- **Activity Type Prediction**

  Mechanism: Multi-Task Extension. We don't just predict one number; we predict 11 activity clusters (Hardware, Calibration, Testing, etc.) simultaneously

  Consistency Loss: The model ensures that the sum of all activities equals the total project cost ($\mathcal{L}_{cons} = |\sum \hat{y_c} - \widehat{y_{total}}|$).

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RQ: 3-6

Methodology

Core Algorithm: Hierarchical Conformal Quantile Ensemble

- **The Hybrid Formula:**

$$\hat{y}_{final} = \alpha \cdot Q_{sizing}(X) + (1-\alpha) \cdot \hat{y}_{ensemble}(X)$$

- **Components:**

  α : Confidence score from the Sizing Classifier.

  Q_{sizing}: Local Quantile Model specialized for the project size (Small/Medium/Large).

  $y_{\widehat{ensemble}}$: Global weighted average of ExtraTrees & Random Forest.

  Why this approach? It dynamically balances local expertise with global stability.

Model Benchmarks

RQ: 5

Model Benchmarking: R&D Effort Prediction (Hours)

•**Evaluation Metric:** The primary validation target was Manpower Hours, as it serves as the core driver for total R&D cost

•**Goodness of Fit:** The proposed HCQE architecture demonstrates superior fit (R^2 =0.92) compared to linear baselines (Linear Regression R^2=0.48), proving that hierarchical modeling is necessary for this dataset

•**Operational Reliability:** In terms of "Ballpark" usability, the model provides a usable estimate (within 30% deviation) in 69.7% of test cases, whereas traditional regression models fail to reach 40% accuracy in this range

Model	Within 30%	Within 50%	MAE (h)	R^2
AdaBoost	60.6%	81.8%	2,035	0.89
Extra Trees	60.6%	75.8%	2,701	0.77
Random Forest	48.5%	72.7%	3,351	0.73
Decision Tree	30.3%	66.7%	3,953	0.74
Ridge (α=10)	24.2%	33.3%	4,717	0.73
SVR (Linear)	24.2%	33.3%	7,733	0.00
Linear Regression	15.2%	39.4%	6,164	0.48
Lasso	15.2%	39.4%	6,156	0.48
HCQE (Ours)	**69.7%**	**81.8%**	**1,974**	**0.92**

RQ: 6

Model Benchmarking: R&D Effort Prediction (Price)

- **Baseline:** Linear Regression (R^2= 0.81) — *Good baseline, stable performance.*
- **Iteration 2:** Random Forest (R^2 = 0.89) — *Tried boosting, but performance dropped slightly.*
- **Final (HCQE):** Ensemble + Decomposition (R^2 = 0.88) — *High accuracy achieved (Ours).*

Model	Within 30%	Within 50%	MAE (K€)	R^2
Linear Regression	39.4%	66.7%	1,344	0.81
Ridge (α=10)	36.4%	63.6%	1,341	0.81
Lasso	39.4%	66.7%	1,336	0.81
SVR (Linear)	42.4%	69.7%	1,125	0.79
Decision Tree	75.8%	84.8%	973	0.76
AdaBoost	45.5%	66.7%	1,491	0.63
Random Forest	48.5%	75.8%	911	0.89
Extra Trees	69.7%	84.8%	797	0.89
HCQE (Ours)	**78.8%**	**84.8%**	**625**	**0.88**

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Comprehensive

Output: The Excel Report

The final output of our AI-Based Ballpark Quotation Tool is a detailed, user-friendly Excel report, designed to provide comprehensive insights briefly.

Executive Summary

Detailed Cost Breakdown

Confidence Intervals

Clearly defined confidence levels for the estimation, providing a realistic range and aiding in strategic decision-making.

Limitations & Future Work

- **Limitations**
  - The system is trained on a limited amount of historical product request data, which may lead to variability in prediction accuracy across different project types.
  - The current feature extraction pipeline relies on a specific LLM; performance may vary when deployed in environments using alternative enterprise LLMs.

- **Future Work**
  - As additional historical data become available, the framework can be retrained to improve accuracy and uncertainty calibration.
  - Future work includes validating the feature extraction pipeline with enterprise-standard LLMs and further integrating the tool into industrial workflows.

Conclusion

- **Objective:**

This project aimed to support early-stage R&D quotation at IVECO Group by enabling faster and more consistent ballpark cost estimations from semi-structured Product Requests.

- **Method:**

A hybrid pipeline combining LLM-based feature extraction, a hierarchical predictive model for effort and cost estimation, and a retrieval-augmented mechanism for contextual justification was designed and evaluated.

- **Effectiveness:**

Experimental results on historical PR–Offer data show that the proposed approach can estimate R&D effort and cost with measurable accuracy while improving consistency and explainability compared to fully manual processes.

THANK YOU

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WE GO BEYOND